

AI Interview Preparation Assistant (AIPA)

Product Requirements Document (PRD) v1.0

Project Overview

AI Interview Preparation Assistant (AIPA) is an AI-powered platform designed to help job seekers prepare for interviews by analyzing their resume against a target job description.

The platform identifies skill gaps, evaluates profile compatibility, generates personalized interview questions, recommends learning paths, and conducts mock interview sessions.

The objective is to provide candidates with actionable feedback and targeted preparation guidance before interviews.

Problem Statement

Job seekers often struggle with:

- Understanding whether their profile matches a job description.
- Identifying missing skills.
- Preparing relevant technical interview questions.
- Preparing behavioral and HR interview questions.
- Creating a focused learning roadmap before interviews.
- Practicing realistic interview scenarios.

Most existing solutions provide generic advice rather than personalized recommendations based on the user's resume and target role.

Product Vision

Enable candidates to receive AI-driven interview preparation tailored to their resume, experience level, and target job role.

The platform should function as an intelligent interview coach capable of:

- Analyzing resumes
- Understanding job requirements
- Identifying strengths and weaknesses
- Generating interview preparation plans
- Simulating interview sessions

Target Users

Primary Users

- Freshers
- Students
- Job Seekers
- Career Switchers

Secondary Users

- Professionals preparing for job changes
 - Recruiters conducting screening exercises
 - Career coaching organizations
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Core Objectives

The system should:

1. Analyze uploaded resumes.
 2. Analyze job descriptions.
 3. Compare candidate profiles against job requirements.
 4. Identify matching and missing skills.
 5. Generate interview questions.
 6. Create personalized learning roadmaps.
 7. Provide mock interview functionality.
 8. Generate downloadable preparation reports.
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Functional Requirements

Module 1: Resume Processing

Description

Allow users to upload resumes.

Supported Formats

- PDF
- DOCX

- TXT

Features

- Resume Upload
- Text Extraction
- Resume Parsing
- Candidate Information Extraction

Extracted Information

- Name
- Email
- Skills
- Experience
- Education
- Certifications
- Projects
- Technologies

Output Example

```
{  
  "name": "John Doe",  
  "experience": "2 Years",  
  "skills": [  
    "Python",  
    "FastAPI",  
    "PostgreSQL"  
  ]  
}
```

Module 2: Job Description Processing

Description

Users can upload or paste a job description.

Features

- JD Upload
- JD Parsing
- Requirement Extraction

Extracted Information

- Required Skills
- Preferred Skills
- Experience Required
- Role Title
- Responsibilities

Output Example

```
{
  "role": "Backend Engineer",
  "required_skills": [
    "Python",
    "Docker",
    "AWS"
  ]
}
```

Module 3: Skill Gap Analysis

Description

Compare resume skills with job requirements.

Outputs

- Matching Skills
- Missing Skills
- Match Percentage
- Candidate Strengths
- Candidate Weaknesses

Example

```
{
  "match_score": 78,
  "matching_skills": [
    "Python",
    "FastAPI"
  ],
  "missing_skills": [
    "Docker",
    "AWS"
  ]
}
```

```
]
}
```

Module 4: Interview Question Generator

Description

Generate interview questions tailored to the candidate and job description.

Categories

Technical Questions

Example:

- Explain FastAPI Dependency Injection.
- What is database indexing?

Scenario Questions

Example:

- Design a scalable notification service.

HR Questions

Example:

- Tell me about yourself.
- Why should we hire you?

Project-Based Questions

Questions generated from projects listed in the resume.

Module 5: Learning Roadmap Generator

Description

Create a personalized learning roadmap based on skill gaps.

Output

Missing Skills:

- Docker
- AWS

Learning Plan:

Week 1:

Docker Fundamentals

Week 2:

Docker Compose

Week 3:

AWS EC2 Basics

Week 4:

AWS Deployment Project

Module 6: Mock Interview Assistant

Description

Interactive AI interviewer.

Features

- Interview Session Creation
- Dynamic Follow-up Questions
- Answer Evaluation
- Improvement Suggestions

Interview Types

- Technical
 - HR
 - Mixed
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Module 7: Report Generator

Description

Generate a complete preparation report.

Report Contents

- Resume Summary
- JD Summary
- Match Score
- Skill Gap Analysis
- Recommended Learning Plan
- Interview Questions
- Preparation Tips

Export Formats

- PDF
 - Markdown
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Non-Functional Requirements

Performance

- Resume Analysis < 30 seconds
- JD Analysis < 15 seconds
- Skill Gap Analysis < 20 seconds
- Report Generation < 60 seconds

Scalability

Support multiple concurrent users.

Reliability

Failed uploads should not impact existing sessions.

Security

- Secure file uploads
- Input validation
- File type validation
- API rate limiting

Technology Stack

Frontend

- React / Next.js
- Tailwind CSS

Backend

- FastAPI
- Pydantic
- SQLAlchemy

Database

- PostgreSQL

AI Layer

- OpenAI
- Ollama
- LangChain

Vector Database

- ChromaDB or Qdrant

Document Processing

- PyPDF
- python-docx

API Endpoints

Resume

POST /api/resume/upload

GET /api/resume/{id}

Job Description

POST /api/job-description/upload

POST /api/job-description/text

Analysis

POST /api/analyze

GET /api/analyze/{id}

Interview Questions

POST /api/questions/generate

Learning Plan

POST /api/roadmap/generate

Mock Interview

POST /api/interview/start

POST /api/interview/answer

GET /api/interview/result

Reports

POST /api/report/generate

GET /api/report/{id}

Success Metrics

The project will be considered successful if:

- Resume parsing accuracy > 90%
 - Skill extraction accuracy > 85%
 - Interview question relevance > 80%
 - Report generation success rate > 95%
 - End-to-end analysis completed within 2 minutes
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Phase 1 Scope (15-Day Internship)

Mandatory Features:

- Resume Upload
- JD Upload
- Resume Parsing
- JD Parsing
- Skill Gap Analysis
- Interview Question Generation
- Learning Roadmap Generation
- FastAPI APIs
- Basic Frontend

Optional Features:

- Mock Interview Chat
- PDF Export
- User Authentication

Out of Scope:

- Multi-user organizations
 - Payment systems
 - Recruiter dashboard
 - Real-time voice interviews
 - Video interview analysis
 - Advanced analytics
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Deliverables

1. Source Code Repository

2. API Documentation
3. Setup Guide
4. Project Architecture Diagram
5. Demo Video
6. Final Presentation

Project Status: Internship Deliverable - Version 1.0